To scale this knowledge for your team, you may wish to implement some tooling in your project to help automate data-level checks. In the next section, you'll use an open source analysis tool that will help you achieve that.

Automating Consistency Checks In Development

To help quickly identify invalid or inconsistent data, you'll need the help of some tooling.

For Rideshare, use the active_record_doctor[19] gem to analyze the database. If you're not familiar with adding Ruby gems to Rails projects, here's a quick refresher. Open the Gemfile and add the gem name to the development group. From your terminal, run bundle install to add it. This gem is already added to Rideshare and you can begin using it.

Active Record Doctor is a command line program. Run bin/rake active_record_doctor from the Rideshare directory in your Terminal.

One of the checks Active Record Doctor performs is to check that Primary Key and Foreign Key columns have matching types.

You may have a Foreign Key column with the integer type and a Primary Key using the bigint type. Change the Foreign Key column to use the bigint type so that they match.

Active Record Doctor also scans your Active Record source code, comparing the Validations with the database configuration and constraints. For example, when a uniqueness model validation exists, but no corresponding UNIQUE constraint exists, the tool suggests adding the UNIQUE constraint. The tool also suggests adding missing Foreign Key constraints.

Another gem to install and experiment with is called database_consistency.[20] Database Consistency makes suggestions by comparing Active Record models and database objects. This gem is also added to Rideshare.

For example, when a LIMIT is specified on a column in the database, but a similar length limit doesn't exist as a Active Record length validation, the tool suggests creating a length validation to match.

Using Active Record Doctor and Database Consistency as part of your Rails project will help improve consistency and correctness in your data and scale this knowledge to your development team.

19. https://github.com/gregnavis/active_record_doctor
20. https://github.com/djezzzl/database_consistency